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A Cross-Species Lexical Repertoire Reveals a Biological Code Linking Acoustic Form to Communicative Meaning

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Abstract

Understanding animal vocalizations means connecting acoustic form to communicative meaning. Years of field studies have produced extensive knowledge of such mappings, yet no unified framework exists for comparing them across species. We introduce AnimalLex, a curated cross-species repertoire database comprising 373 calls from 57 species spanning mammals, birds, and amphibians, annotated with standardized semantic and acoustic descriptions synthesized from the scientific literature and validated by domain experts. Using AnimalLex, we characterize what animals talk about and how. We find that acoustic and semantic distances are significantly correlated within species, across related species, and even across taxa—evidence for a conserved form-to-meaning structure that transcends phylogenetic boundaries. We leverage this structure to perform cross-species lexical translation: given any call, we can identify its closest analogue in any other species. These results suggest that a biological code linking acoustic form to communicative function is a deep organizational principle of animal communication, and that large-scale repertoire curation—now made practical by language models—unlocks new possibilities for comparative and translational bioacoustics. An interactive tool is provided to explore the database and perform translations at scale.

1 Introduction

Decades of field research have established that animal vocalizations are not simple reflexes but structured communicative signals whose forms are shaped by function Marler, 1955. Alarm calls are short, broadband, and easily localized—suited for rapid threat broadcast. Contact calls are tonal and repetitive—suited for long-distance propagation and individual recognition. Infant-directed calls are high-pitched across taxa—plausibly linked to shared attentional mechanisms. Yet these patterns have been documented one species at a time, and no synthesis has asked whether they constitute a universal code: a systematic, learnable relationship between acoustic form and communicative meaning that holds across the tree of life.

The question is not merely empirical but conceptual. In linguistics, the arbitrariness of the sign—the principle that a word's sound bears no necessary relation to its meaning—has been a foundational tenet since Saussure Saussure, 1916. Counter-evidence is accumulating: sound symbolism and crosslinguistic iconicity demonstrate that form-to-meaning mappings

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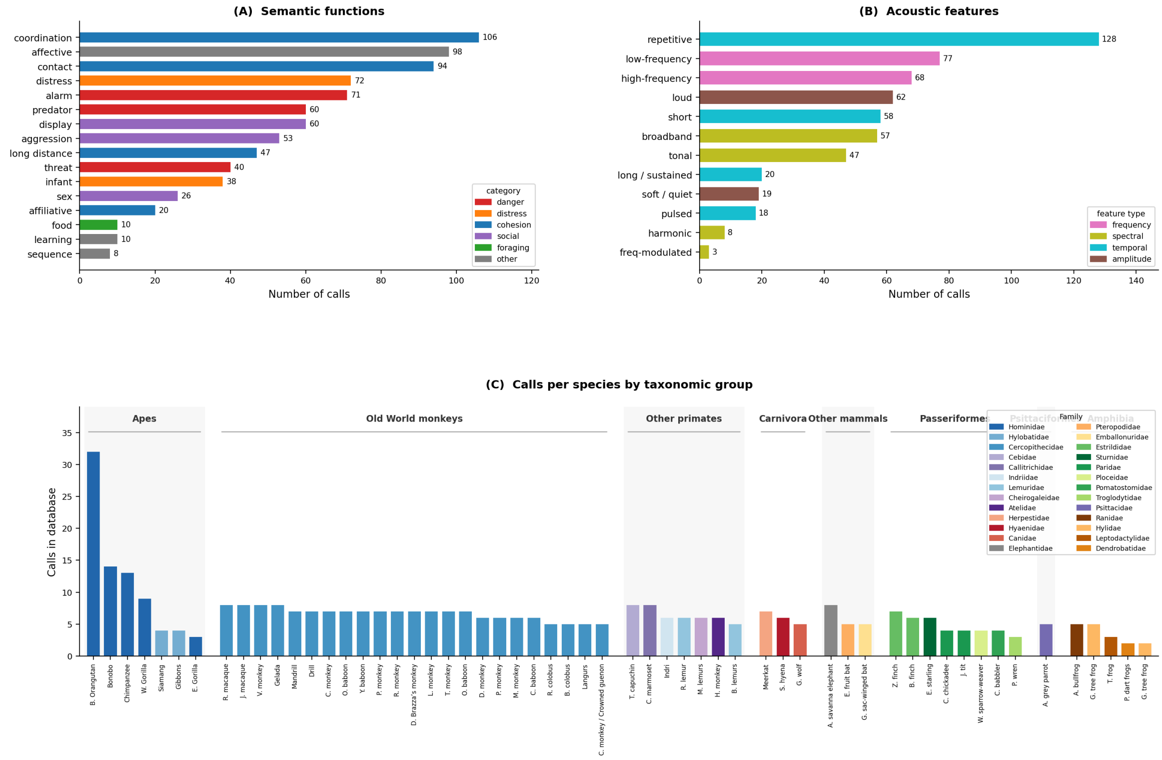

Figure 1. **AnimalLex: a curated cross-species lexical repertoire. (A)** Frequency of semantic ontology keywords across all 373 calls, colored by communicative category. Coordination, contact, alarm, and affective signaling dominate; rarer categories include referential signaling, individual identity, and learning-related calls. **(B)** Frequency of acoustic features extracted by term-matching from free-text descriptions. Tonal, high-frequency, and repetitive features are most common; soft/quiet and harmonic features are rarer. **(C)** Calls per species, grouped by taxonomic group (Apes, Old World monkeys, other primates, Carnivora, other mammals, Passeriformes, Psittaciformes, Amphibia) and colored by family. Within each group, species are sorted by repertoire size. Repertoire sizes range from 2 to 32; the largest is the Bornean orangutan (32 calls), reflecting the depth of primary literature on great apes.

exist and are learned in human language Dingemanse et al., 2015. In non-human animals the question is more open still. If such a biological code exists, it could arise through two mechanisms: convergent evolution driven by shared physical and ecological constraints (e.g., the localizability demands of alarm signaling), or phylogenetic conservation of ancestral form-to-meaning associations within lineages. These mechanisms make different predictions—the former produces cross-taxon universals, the latter family-level regularities—and can in principle be distinguished with sufficient data.

Here we test for a biological code at scale by assembling AnimalLex, a curated database of 373 calls from 57 species (Fig. 1). Each entry pairs a free-text acoustic description with a free-text semantic description and standardized ontology keywords, drawn from peer-reviewed literature and validated by domain experts. We embed both descriptions independently into continuous vector spaces and ask whether their pairwise distance structures are aligned—within species, across species, and across higher taxa. We find that they are: acoustic distance is a significant predictor of semantic distance at every taxonomic level tested. A simple linear model trained on this alignment generalizes across species and taxa, demonstrating that the code is learnable. We apply it directly to perform cross-species lexical translation—mapping any documented call to its closest human-language analogue, and any human concept to its closest animal vocalization—and provide an interactive web application for broad exploration.

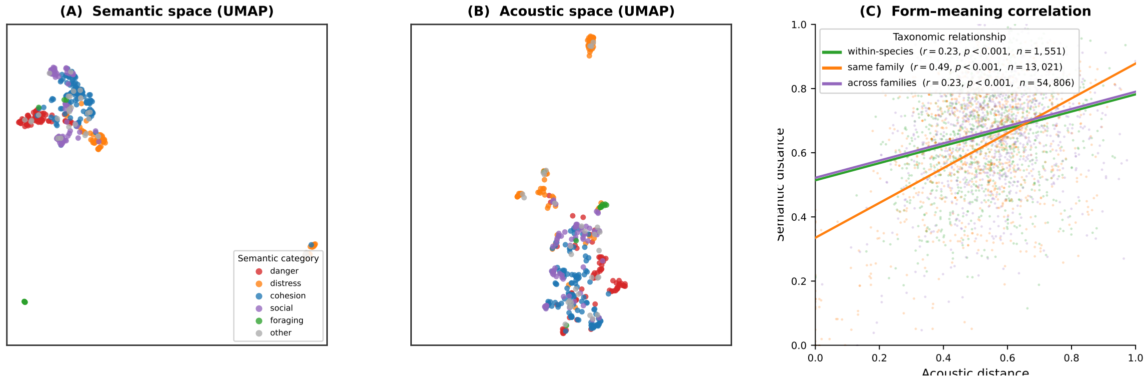

Figure 2. **Alignment of acoustic and semantic embedding spaces. (A)** Semantic UMAP: each point is a call, colored by communicative category (legend inside panel). Alarm/threat calls (red) cluster together; contact/cohesion calls (blue) form a separate region. **(B)** Acoustic UMAP: the same calls positioned by their acoustic description embedding. The topological similarity to (A)—shared cluster identities despite independent embeddings—is the visual signature of the biological code. **(C)** Mantel scatter: 1,000 randomly sampled pairs per group, pairwise acoustic distance vs. semantic distance, colored by taxonomic relationship. r values from permutation Mantel tests on all pairs; same-family cross-species pairs ($r = 0.49$) show the strongest alignment.

2 A cross-species lexical repertoire

AnimalLex was assembled by prompting a large language model (GPT-4o) to structure knowledge from peer-reviewed literature for each target species, producing entries that record call name, free-text acoustic description, free-text semantic description, standardized ontology keywords, subjective reliability, and primary references. Entries were then reviewed and corrected by domain experts (see Methods). The 57 species span three vertebrate classes: 43 mammals, 9 birds, and 5 amphibians, covering taxa from African savanna elephants (*Loxodonta africana*) and gray wolves (*Canis lupus*) to Carolina chickadees (*Poecile carolinensis*) and túngara frogs (*Engystomops pustulosus*).

The most common semantic categories are coordination ($n = 106$), affective signaling ($n = 98$), contact and cohesion ($n = 94$), distress ($n = 72$), and alarm ($n = 71$; Fig. 1A). Less frequent but theoretically important categories include referential signals ($n = 5$), individual-identity calls ($n = 8$), and learning-associated vocalizations ($n = 10$). On the acoustic side, descriptions span the full range of spectro-temporal parameters: tonal vs. broadband energy, frequency range and modulation, temporal patterning (pulsed, repetitive, graded), and call-internal structure (notes, syllables, phrases; Fig. 1B). Together these two annotation axes provide the substrate for testing form-to-meaning alignment.

3 Acoustic and semantic spaces are aligned across the tree of life

We embedded each call's acoustic and semantic descriptions separately into a shared sentence-transformer space (see Methods) and computed pairwise cosine similarities across all $373(373-1)/2 = 69{,}378$ call pairs. A Mantel test Mantel, 1967 of the correlation between acoustic and semantic similarity matrices yields $r = 0.31$, $p < 0.001$ (9,999 permutations; Fig. 2C). A partial Mantel test controlling for species identity yields an identical coefficient ($r = 0.31$, $p < 0.001$), confirming that the correlation is not driven by the trivial within-species similarity structure: calls from the same species are not the reason acoustic and semantic spaces are aligned.

Examining subsets of pairs by taxonomic relationship reveals additional structure. Pairs from the same family but different species show the strongest correlation ($r = 0.49$, $p < 0.001$, $n = 13{,}021$ pairs), while within-species pairs (pooled across all species; $r = 0.23$, $p < 0.001$, $n = 1{,}551$) and cross-family pairs ($r = 0.23$, $p < 0.001$, $n = 54{,}806$) are weaker and similar to each other. The same-family peak suggests that acoustic-semantic alignment is most visible when species share evolutionary history but have begun to acoustically diverge—consistent with a code that is phylogenetically conserved while acoustic surface form drifts.

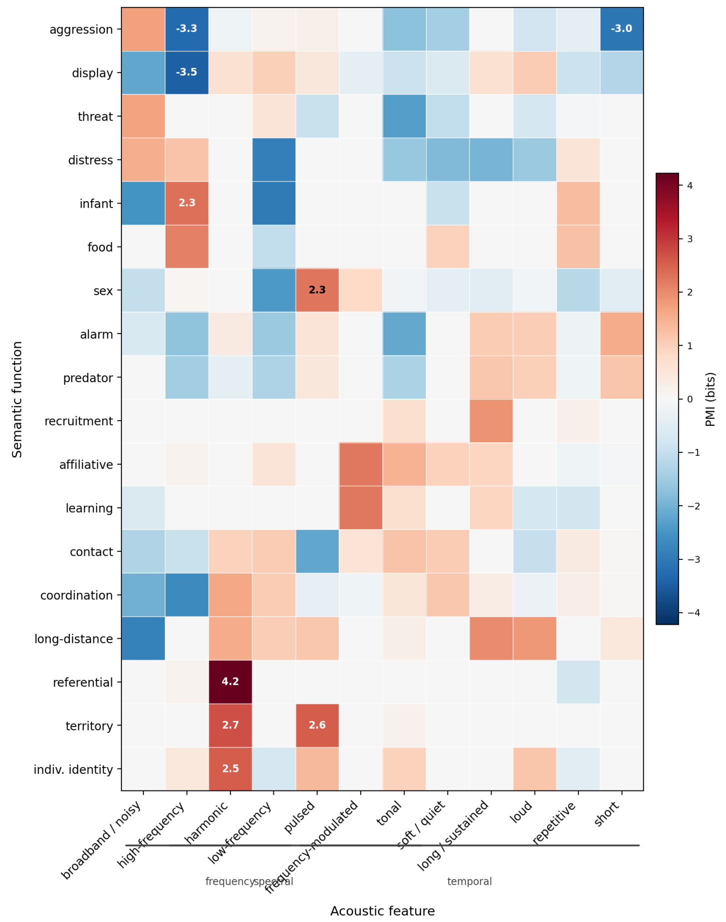

Figure 3. **Keyword-level structure of the biological code.** Pointwise mutual information (PMI, bits) between semantic functions (rows) and acoustic features (columns), estimated across all 373 calls in AnimalLex. Rows and columns are ordered by hierarchical clustering; numbers annotate the six strongest and three weakest cells. Warm colors = acoustic feature over-represented in calls with that semantic function; cool colors = under-represented. Three acoustic feature groups are bracketed on the x-axis.

To decompose the signal into interpretable units, we computed the pointwise mutual information (PMI) between acoustic features and semantic keywords across all calls (Fig. 3). Several strong, interpretable associations emerge. High-frequency calls are over-represented in infant-directed and food contexts (PMI $\approx$ 2.1–2.3) and sharply under-represented in display and aggression contexts (PMI ≈ -3.3 to -3.4). Broadband and noisy calls are enriched for aggression and threat semantics (PMI $\approx$ 1.7), consistent with the "harsh = aggressive" frequency code Morton, 1977. Long/sustained and loud features co-occur with long-distance cohesion calls (PMI $\approx$ 1.8–2.0), consistent with propagation demands. These keyword-level associations provide a transparent, mechanistic decomposition of the biological code and replicate functional acoustic ecology predictions across 57 species simultaneously.

4 Form-to-meaning mapping is learnable and generalizes across taxa

The previous section demonstrates that there is stability in the form-to-meaning mapping. We trained models to exploit this regularity: predicting semantic embeddings from acoustic embeddings under three hold-out conditions of increasing ambition (Table 1). Within-repertoire, ridge regression achieves a mean cosine similarity of 0.74 and MRR of 0.46—substantially above the random baseline (cos. 0.36, MRR 0.11). Generalization to held-out species is equally strong (cos. 0.74, MRR 0.80), and remains well above chance even when entire families are held out (cos. 0.68, MRR 0.70). The retrieval baseline approaches ridge regression in MRR for held-out

Table 1. **Cross-generalization of acoustic-to-semantic prediction.** Four models are evaluated under three hold-out conditions of increasing difficulty. *Cos.* = mean cosine similarity between predicted and true semantic embedding (higher is better, max 1). *MRR* = mean reciprocal rank of the true call among all test-set calls ranked by similarity to the prediction (higher is better, max 1). The random baseline shuffles acoustic–semantic pairings within each test set. Retrieval ($k = 1$) returns the nearest neighbour in acoustic space among training calls. Within-repertoire values are means over 10 random folds; held-out species and family values are means over leave-one-out folds (55 and 26 folds respectively).

	Within repertoire		**Held-out species**		**Held-out family**	
Model	Cos.	MRR	Cos.	MRR	Cos.	MRR
Random	0.36	0.11	0.46	0.41	0.47	0.42
Retrieval ($k = 1$)	0.67	**0.51**	0.68	0.78	0.60	0.67
Linear (ridge)	**0.74**	0.46	**0.74**	**0.80**	**0.68**	**0.70**
MLP	0.71	0.45	0.73	**0.80**	0.66	0.68

conditions, indicating that the acoustic-to-semantic mapping is largely a global alignment of the two spaces rather than a complex nonlinear transformation. The MLP offers no consistent advantage over ridge regression, consistent with this interpretation. Together these results show that the biological code identified by the Mantel analysis is not an artefact of within-species structure: it generalises to species and entire families unseen at training time. In other words, our existing knowledge of species transfers to new species.

5 Lexical translation across species

The form-to-meaning structure identified above enables a direct translational application that operates entirely within the animal kingdom. Given a call from species A, we retrieve the call in species B whose semantic embedding is nearest—a lexical translation from one repertoire to another. When both species are represented in AnimalLex with full semantic annotations, the mapping is performed in semantic space by retrieval. When semantic annotations are absent for one or both species—as will routinely be the case for poorly studied taxa—the acoustic-to-semantic predictor from the previous section supplies a best-guess semantic embedding from acoustic features alone, enabling translation even across the boundary of documented knowledge.

This two-mode pipeline (retrieval for annotated calls; prediction for unannotated ones) makes AnimalLex a living resource: every new species added expands the reach of translation, and the predictor provides a scaffold for annotating the next. We release an interactive web application ([placeholderURL]) in which users can select any call and browse its nearest semantic neighbours across all species in AnimalLex, with the option to include predicted-semantics calls from unannotated recordings (see Appendix A).

A natural extension is to learn the translation *directly in acoustic space*, bypassing the semantic intermediate. Two species that are phylogenetically close should have nearly aligned acoustic spaces—similar vocal anatomies, shared evolutionary pressures, conserved call repertoires—and their repertoires should be mappable by a small rotation. Phylogenetically distant species may require a larger transformation. This suggests learning a family of species-pair rotations R_{AB} (Procrustes-style, constrained to be orthogonal) regularised by phylogenetic distance: $\|R_{AB} - \mathbf{I}\|_F \leq \lambda \cdot d_{\text{phylo}}(A, B)$, where d_{phylo} is branch length in a dated phylogeny and λ controls how much divergence is allowed per unit evolutionary distance. Such a model would be fully audio-grounded, require no semantic labels, and could scale to the thousands of species for which recordings but no annotations exist. We leave this to future work, but note that the form-to-meaning results above provide the theoretical justification: because acoustic and semantic spaces are aligned, a rotation in acoustic space is also, approximately, a rotation in semantic space.

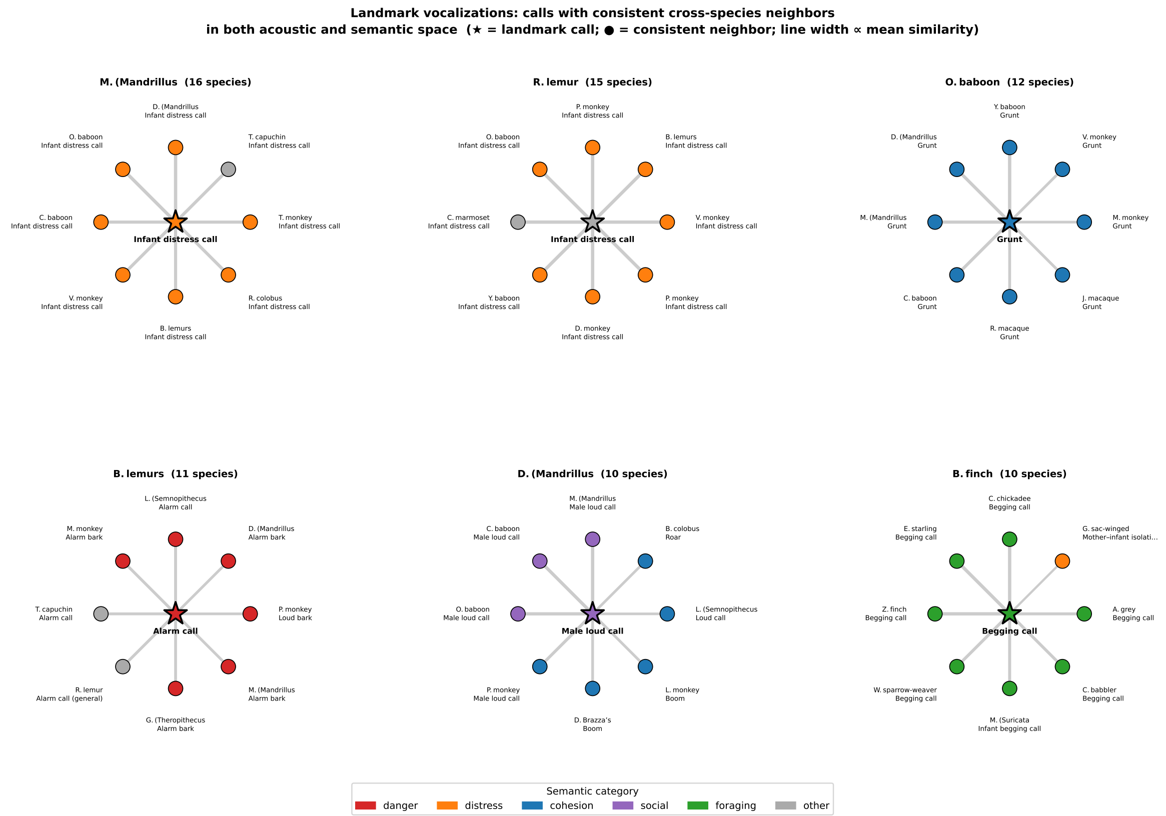

Figure 4. **Cross-species lexical translation. Calls with consistent cross-species neighbors in both acoustic and semantic space.** Each rosette shows one landmark call (★, labeled with its call name at center) connected to up to eight calls from other species whose acoustic *and* semantic embeddings both rank among its top-20 nearest neighbors. Node color encodes semantic category (same scheme as Fig. 2); spoke width is proportional to mean acoustic–semantic similarity. Parenthetical counts give the number of distinct species with at least one consistent neighbor (range: 10–16). These landmark calls represent acoustic-semantic motifs independently co-opted across the tree of life—the concrete instances of the biological code quantified by the Mantel test (Fig. 2C).

6 Discussion

Our results establish three things. First, animal vocalizations carry systematic form-to-meaning relationships detectable across the tree of life—not just within well-studied model systems. Second, these relationships are learnable and generalizable: a model trained on some species predicts communicative meaning from acoustics for entirely held-out species and families. Third, this learnability enables practical cross-species lexical translation—finding, for any call, its closest semantic analogue in any other repertoire—including for species whose calls have never been semantically annotated.

Two non-exclusive mechanisms can explain the cross-taxon alignment. Convergent functional pressures may independently channel evolution toward similar acoustic solutions for similar communicative problems: the physics of sound localization favors broadband calls for alarm regardless of ancestry Marler, 1955; the acoustics of long-range propagation favor tonality for cohesion calls Morton, 1977. Phylogenetic inertia may additionally preserve ancestral form-to-meaning associations within lineages, explaining why correlations are strongest within species and weaken—without vanishing—across families. These mechanisms predict different patterns in phylogenetically controlled analyses, which future work with broader taxonomic sampling can test.

Several caveats apply. AnimalLex represents the current state of published knowledge, which skews toward well-studied mammals; birds and amphibians—acoustically diverse and experimentally tractable—are underrepresented. Our acoustic embeddings are derived from textual descriptions, not from audio directly; future work integrating audio representations Hagiwara, 2022; Robinson et al., 2025 will test whether the form-to-meaning alignment is stronger in the raw signal domain. Lexical translation should be understood as semantic proximity within the space of documented ethological descriptions—not as a claim about subjective experience, cognitive content, or the *Umwelt* of the signaling animal. Any mapping through human language (see Appendix B) inherits an additional anthropomorphic bias and should be treated as a heuristic rather than a literal gloss.

Despite these limitations, the demonstration that form-to-meaning structure is learnable and cross-taxon generalizable opens a practical path forward: with each new species added to AnimalLex, the translation model improves, and with each improvement the model can bootstrap annotation of the next species. This virtuous cycle—human knowledge, compressed by language models, refined by acoustic-semantic prediction, extended to undocumented taxa—is the long-term promise of the approach introduced here.

7 Methods

7.1 Database construction

AnimalLex was assembled in two stages. We first prompted GPT-4o OpenAI, 2024 with a structured schema specifying six required fields per call: (1) call name, (2) free-text acoustic description (spectro-temporal features, amplitude envelope, call-internal structure), (3) free-text semantic description (communicative function, behavioral context, receiver responses), (4) ontology keywords drawn from a fixed controlled vocabulary, (5) subjective reliability rating (high / medium / low / contested), and (6) URLs to primary-literature references. The model was instructed to restrict outputs to findings supported by peer-reviewed sources and to flag contested or uncertain claims. In the second stage, domain experts in avian, mammalian, and amphibian bioacoustics reviewed all entries, corrected factual errors, added missing calls, and updated reliability ratings. Calls rated "low" or "contested" are retained in AnimalLex but excluded from quantitative analyses reported here.

7.2 Embeddings

Acoustic and semantic descriptions were embedded independently using the all-MiniLM-L6-v2 sentence transformer Reimers and Gurevych, 2019, which maps variable-length text to a 384-dimensional vector space trained on semantic similarity tasks. Embeddings were ℓ_2-normalized before computing cosine distances. We deliberately used textual embeddings rather than audio-derived representations to ensure that acoustic and semantic descriptions were embedded in a comparable space; the validity of this choice is discussed above and tested in Supplementary C.

7.3 Mantel test

For each subset of calls (all pairs; within-species pairs; within-family pairs; cross-family pairs), we computed the Pearson correlation between the upper triangle of the pairwise acoustic distance matrix and the upper triangle of the pairwise semantic distance matrix. Statistical significance was assessed by permuting the rows and columns of the acoustic distance matrix 10,000 times and computing the fraction of permuted correlations exceeding the observed value.

7.4 PMI analysis

Acoustic keywords were extracted from free-text acoustic descriptions by matching a fixed lexicon of 20 spectro-temporal terms (e.g., *tonal*, *broadband*, *high-frequency*). Semantic keywords were taken directly from the ontology keyword field. For each acoustic–semantic keyword pair (a, s), PMI was estimated as $\log_2 \hat{p}(a, s) - \log_2 \hat{p}(a) - \log_2 \hat{p}(s)$, where probabilities were estimated from call frequencies. Pairs with fewer than three co-occurrences were excluded.

7.5 Acoustic-to-semantic prediction

We trained a ridge regression mapping acoustic embeddings $\mathbf{x} \in \mathbb{R}^{384}$ to semantic embeddings $\mathbf{y} \in \mathbb{R}^{384}$, regularized with ℓ_2 penalty λ selected by leave-one-species-out cross-validation. Predicted embeddings were ℓ_2-normalized before evaluation. Performance was measured by (i) mean cosine similarity between predicted and true semantic embeddings and (ii) mean reciprocal rank (MRR) of the true call's semantic embedding among all held-out calls, ranked by cosine similarity to the prediction. We additionally evaluated a shallow MLP (two hidden layers of 512 units, ReLU activations, dropout 0.2) and a retrieval baseline (k-nearest neighbors in acoustic space, semantic embedding averaged over the k neighbors). Cross-validation used 10 folds stratified at the species level.

8 Contributions

Conceptualization: A.B., C.D. **Methodology:** A.B., C.D., E.F. **Software:** E.F., G.H. **Validation:** C.D., E.F. **Formal analysis:** A.B., E.F. **Investigation:** A.B., C.D. **Resources:** I.J. **Data curation:** A.B., C.D., E.F. **Writing – original draft:** A.B. **Writing – review & editing:** A.B., C.D., E.F., I.J. **Visualization:** E.F., G.H. **Supervision:** I.J. **Project administration:** I.J. **Funding acquisition:** I.J.

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A Web application

We release an interactive web application at [placeholderURL] that allows users to: (i) browse AnimalLex filtered by species, taxon, semantic category, or acoustic keyword; (ii) translate any call to its nearest semantic neighbour in any other species; (iii) explore semantic and acoustic embedding spaces as interactive 3-D UMAP projections; and (iv) optionally include predicted-semantics calls from unannotated recordings. The application is built on [placeholder: framework] and is freely accessible without registration.

B Human-language translation (anthropomorphic, use with caution)

The translation framework can be applied with human language as one of the two "species": given a word or phrase, embed it in the shared semantic space and retrieve the animal call with the nearest embedding; conversely, given any call, retrieve the closest human-language gloss. This mapping is *not* a claim that animals share human concepts—it reflects only that the ethological descriptions in AnimalLex, written by human researchers, overlap in embedding space with human vocabulary. Every translation step through human language imposes an anthropomorphic frame and risks projecting a human *Umwelt* onto species whose perceptual and cognitive worlds may be radically different. We include it here as an exploratory tool for human audiences, not as a scientific claim.

Table 2. **Human-language translation: selected examples (see caveats above).** For each human phrase, the best-matching call in AnimalLex by semantic embedding similarity. [Placeholder: to be filled once translation pipeline is run.]

Human phrase	Sim.	Best-match call	Acoustic description
danger	—	[species] / [call]	[acoustic]
I am hungry	—	[species] / [call]	[acoustic]
come here	—	[species] / [call]	[acoustic]
I am here	—	[species] / [call]	[acoustic]
stay away	—	[species] / [call]	[acoustic]
greeting	—	[species] / [call]	[acoustic]
let's move together	—	[species] / [call]	[acoustic]
I am afraid	—	[species] / [call]	[acoustic]
this is mine	—	[species] / [call]	[acoustic]
help me	—	[species] / [call]	[acoustic]

C Validation against audio representations

[Placeholder: comparison of Mantel test results when acoustic embeddings are derived from (a) text descriptions, as in the main analysis, vs. (b) audio features extracted by AVES Hagiwara, 2022 or NatureLM-audio Robinson et al., 2025 for the subset of calls for which verified audio samples are available. This analysis tests whether the form-to-meaning alignment detected through textual descriptions is an artifact of description style or reflects genuine acoustic structure.]

D Full database statistics

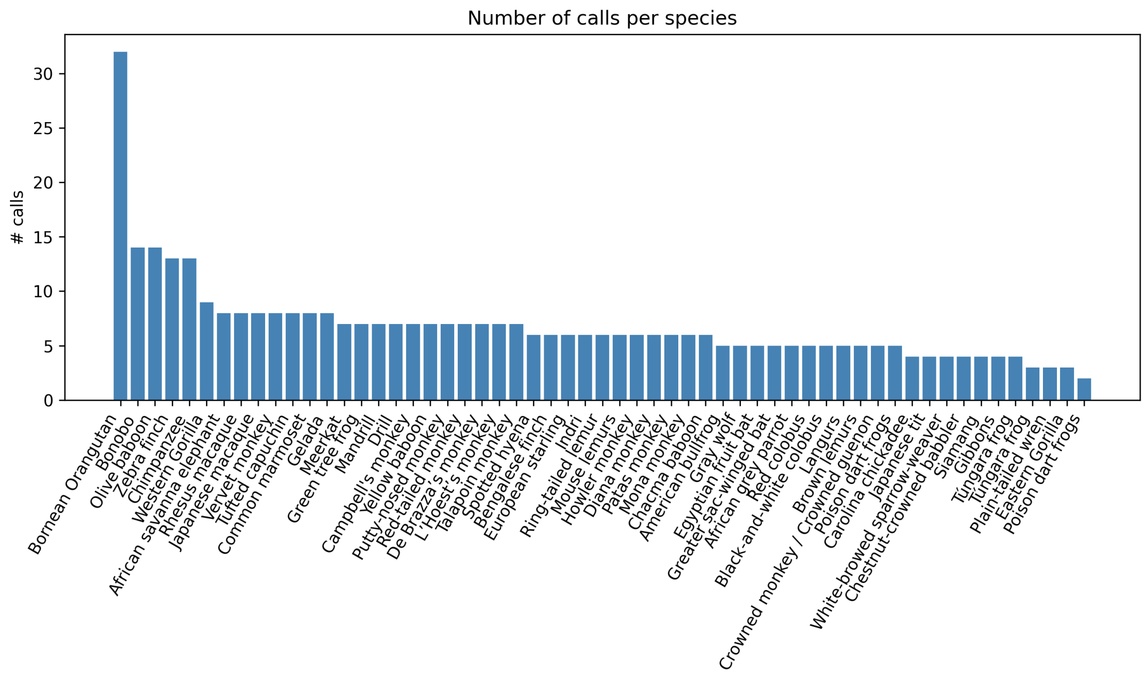

Figure 5. **Calls per species in AnimalLex**, sorted by count. Color encodes taxonomic class. The median repertoire size is [X] calls; the largest repertoire (*Loxodonta africana*) contains [X] calls.

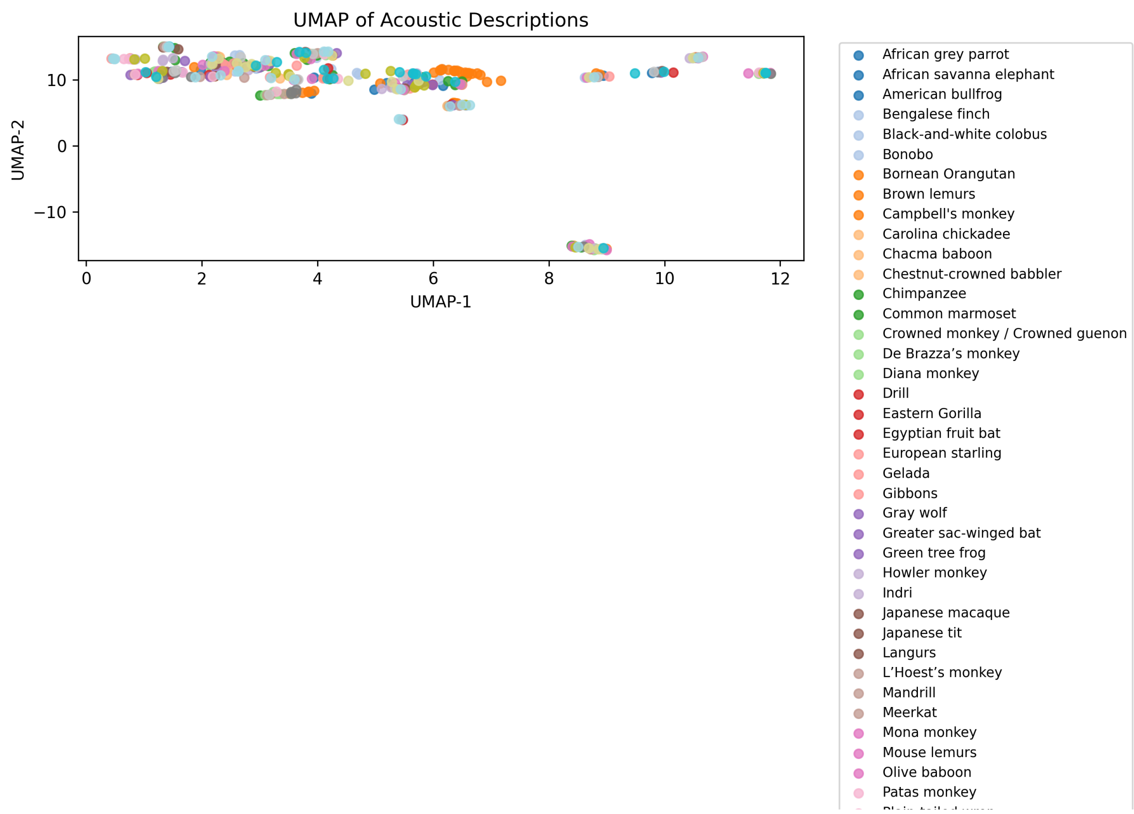

Figure 6. **Acoustic embedding space (UMAP).** Each point is a call in AnimalLex, positioned by acoustic description embedding and colored by taxonomic class. Clusters correspond to spectro-temporal call types (tonal long calls, broadband alarm bursts, etc.) that recur across classes, illustrating the convergent acoustic solutions captured by the database.

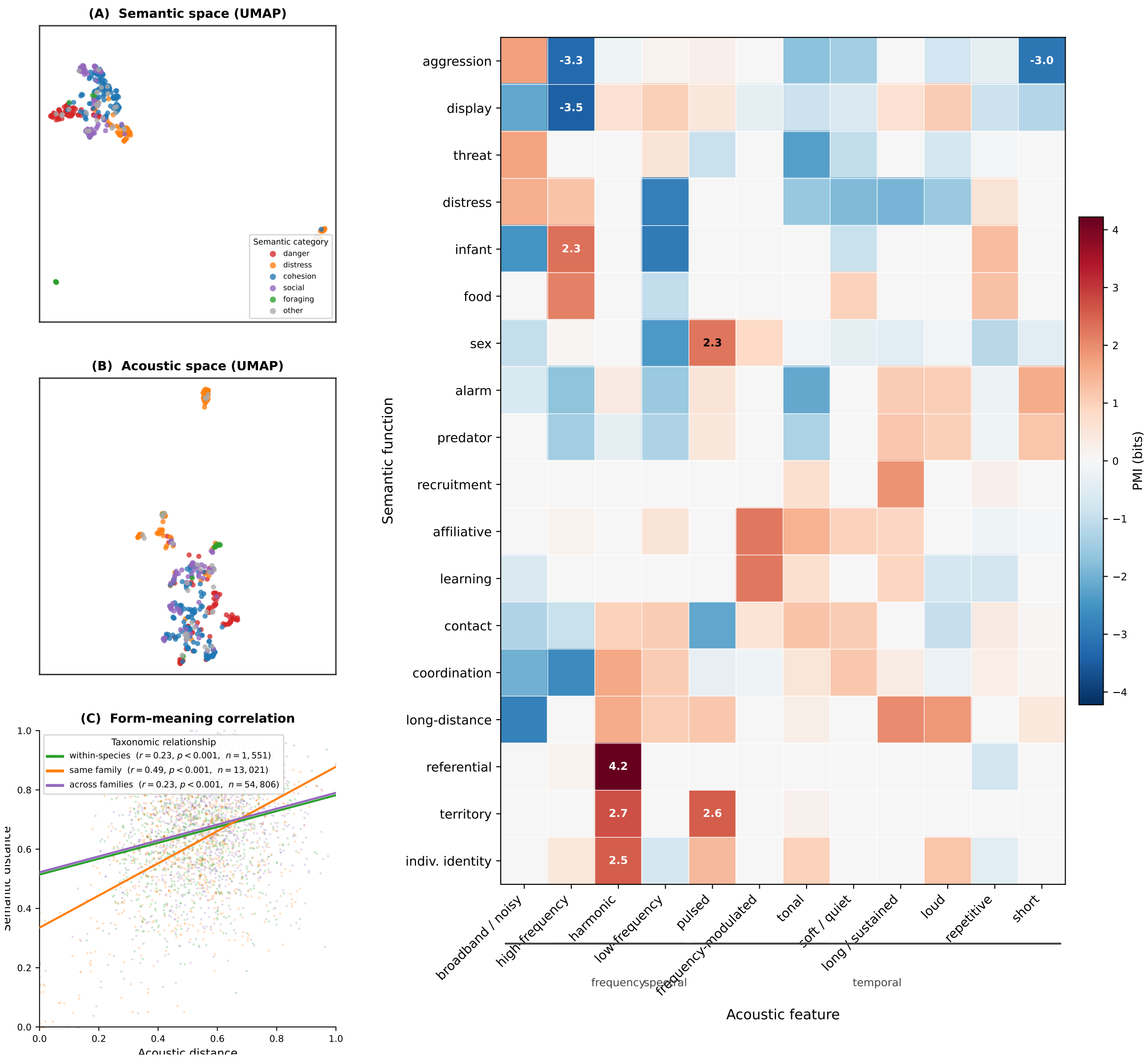

Figure 7. **[Draft] Merged Figs. 2+3 candidate.** Left column: **(A)** semantic UMAP, **(B)** acoustic UMAP, **(C)** Mantel scatter (same as Fig. 2). Right column: **(D)** PMI heatmap with semantic functions as rows and acoustic features as columns (square cells); rows and columns ordered by hierarchical clustering (same data as Fig. 3).